

The SOT Calkin Quotient Topology is Indiscrete

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Status

This packet gives a partial answer to a question in Talponen's *Convex-transitivity of Banach algebras via ideals*, arXiv:1106.1292 [1]. The source asks whether

$$\mathcal{C}(L^\infty) = \mathcal{B}(L^\infty)/\mathcal{K}(L^\infty)$$

is “(SOT-)convex-transitive.” The result below settles the SOT part affirmatively, but in a degenerate topological way: the SOT quotient topology on every nonzero Calkin quotient is indiscrete. The norm convex-transitivity question is not settled.

Source Question

The source paper defines quotient topologies as follows. If X is a Banach space with a locally convex topology τ and $Y \subset X$ is a subspace, then X/Y is considered with the quotient topology whose neighborhoods are sets $U + Y$, even if Y is not τ -closed. It also defines τ -convex-transitivity by requiring

$$\overline{\text{conv}}^\tau(\mathcal{G}(x)) = B_X \quad (x \in S_X).$$

After proving that certain ideals $\mathcal{A}/\mathcal{I}$ are SOT-convex-transitive, the source asks:

$M_{\tilde{\mathcal{P}}}(i, 1)$. In general, we define $N(i, j, m) = 1$ for $\sum_{l=1}^{j-1} |k_{i,l}| < m \leq \sum_{l=1}^j |k_{i,l}|$ and 0 for other values of m . Then the approximation for $T_{\tilde{\mathcal{P}}}$ is obtained as the average

$$x \mapsto \frac{1}{n} \sum_{m=1}^n \sum_{i,j} N(i, j, m) \mathbb{E}(x | C_j) 1_{C_i}, \quad x \in L^\infty$$

of operators, which can be realized in the form (3.8).

We conclude that $T \in \overline{\text{conv}}^{\text{SOT}}(RS : R)$. Since the rotations R applied preserve $\mathcal{I}$ and $\mathcal{A}$, this means that $\mathcal{A}/\mathcal{I}$ is SOT-convex-transitive. $\square$

We do not know if the Calkin algebra $\mathcal{C}(L^\infty) = \mathcal{B}(L^\infty)/\mathcal{K}(L^\infty)$ is (SOT-)convex-transitive. We note that the above ideals $\mathcal{A}$ and $\mathcal{I}$ are in fact u -ideals, as the

Figure 1: The Calkin-algebra question on page 13 of arXiv:1106.1292.

In text form, the relevant sentence is that the author does not know whether the Calkin algebra

$$\mathcal{C}(L^\infty) = \mathcal{B}(L^\infty)/\mathcal{K}(L^\infty)$$

is (SOT-)convex-transitive.

Proof Intuition

The strong operator topology is too weak to see the Calkin quotient. On any Banach space, finite-rank operators are SOT-dense in all bounded operators: to match an operator T on finitely many test vectors, project the finite dimensional span of the corresponding images back onto itself. Since finite-rank operators are compact, the compact ideal is SOT-dense in $\mathcal{B}(X)$. A quotient by a dense subspace has the indiscrete quotient topology. Thus every nonempty set is dense, including every convex hull of an isometry orbit.

Lemma 1. *Let X be a Banach space. The finite-rank operators $\mathcal{F}(X)$ are dense in $\mathcal{B}(X)$ for the strong operator topology.*

Proof. Let $T \in \mathcal{B}(X)$, let $x_1, \dots, x_n \in X$, and let a basic SOT neighborhood of T be prescribed by these vectors and a tolerance $\varepsilon > 0$. Put

$$E = \text{span}\{Tx_1, \dots, Tx_n\}.$$

Since E is finite dimensional, there is a bounded projection $P : X \rightarrow E$: choose a basis of E , extend the coordinate functionals by Hahn–Banach, and assemble the associated finite-rank projection. Then PT has finite rank and

$$(PT)x_i = Tx_i \quad (1 \leq i \leq n).$$

Thus PT belongs to the prescribed SOT neighborhood of T . □

Theorem 1. *Let X be an infinite-dimensional Banach space. Give $\mathcal{B}(X)/\mathcal{K}(X)$ the quotient topology induced by the strong operator topology on $\mathcal{B}(X)$, in the sense used in [1]. Then this quotient topology is indiscrete. Consequently, $\mathcal{B}(X)/\mathcal{K}(X)$ is SOT-convex-transitive whenever it is nonzero.*

Proof. By the lemma,

$$\mathcal{F}(X) \subset \mathcal{K}(X)$$

is SOT-dense in $\mathcal{B}(X)$, so $\mathcal{K}(X)$ is SOT-dense in $\mathcal{B}(X)$. Let $q : \mathcal{B}(X) \rightarrow \mathcal{B}(X)/\mathcal{K}(X)$ be the quotient map. We show that every nonempty open set in the quotient is the whole quotient.

Indeed, a nonempty quotient-open set has the form $q(U)$, where U is a nonempty SOT-open subset of $\mathcal{B}(X)$. Fix any $S \in \mathcal{B}(X)$. Since $\mathcal{K}(X)$ is SOT-dense, the nonempty SOT-open set $U - S$ intersects $\mathcal{K}(X)$. Choose $K_0 \in \mathcal{K}(X) \cap (U - S)$. Then $S + K_0 \in U$, and hence

$$q(S) = q(S + K_0) \in q(U).$$

Since S was arbitrary, $q(U) = \mathcal{B}(X)/\mathcal{K}(X)$. Thus the quotient topology is indiscrete.

Now suppose the quotient is nonzero and take z in its unit sphere. The isometry orbit $\mathcal{G}(z)$ is nonempty, because it contains z under the identity isometry. Hence $\text{conv}(\mathcal{G}(z))$ is a nonempty subset of an indiscrete topological vector space, and its closure is the whole quotient. In particular, it contains the closed unit ball. This is exactly SOT-convex-transitivity in the source’s formal topological sense. □

Corollary 1. *The Calkin algebra*

$$\mathcal{C}(L^\infty) = \mathcal{B}(L^\infty)/\mathcal{K}(L^\infty)$$

is SOT-convex-transitive with the quotient SOT convention of [1].

Proof. Apply the theorem with $X = L^\infty$. Since L^∞ is infinite dimensional, its identity operator is not compact, so the Calkin quotient is nonzero. □

Verification Notes

The proof uses only the source paper’s own convention for quotient topologies and the elementary fact that finite-dimensional subspaces of Banach spaces are complemented. No approximation property of X is required.

The result is not specific to L^∞ ; it applies to every nonzero Calkin quotient $\mathcal{B}(X)/\mathcal{K}(X)$ when the quotient is equipped with the raw SOT quotient topology.

Limitations

This is a partial answer to the printed question. It answers the parenthetical SOT version, but it does not say whether $\mathcal{C}(L^\infty)$ is convex-transitive for the quotient norm. That norm problem remains the substantive open part.

If the intended SOT formulation were meant to pass to a Hausdorff reflection, or to use a topology in which $\mathcal{K}(L^\infty)$ is not dense, then the above argument would identify a mismatch in terminology rather than solve that amended problem.

Novelty and Search Bounds

The local run indexes were searched for arXiv:1106.1292, Calkin algebra, SOT-convex-transitivity, finite-rank SOT density, compact-operator SOT density, and quotient-topology indiscreteness. No duplicate packet was found.

A bounded web search on June 23, 2026 for exact phrases from the source question, the title, and combinations of “Calkin”, “SOT”, and “convex-transitive” found the source paper but no separate answer to this SOT quotient-topology observation. The result should therefore be reviewed as a likely new clarification/partial answer, not as a known literature retrieval.

Human Review Recommendation

Review the packet as a topological clarification. The proof is elementary; the main review question is interpretive: whether the source’s parenthetical SOT question intended the raw quotient topology defined in the preliminaries. If yes, the SOT half is settled and degenerate. The norm convex-transitivity of $\mathcal{C}(L^\infty)$ remains open.

References

- [1] J. Talponen, *Convex-transitivity of Banach algebras via ideals*, arXiv:1106.1292.